SAMPLE

Earth's Sky: The Seasons

Solstices, Equinoxes, and Why Earth Tilts

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Introduction

Every child learns the sequence, spring, summer, autumn, winter, almost before learning to read, and the learning feels like a fact of the world, like the sky being blue. Yet the seasons are not a fact of the world but a consequence of one, and the consequence is stranger than most people ever learn. The Earth's year, the planet's orbit around the Sun, is nearly circular; the planet's distance from its star changes through the year by only a few percent, and the planet is actually farthest from the Sun, the whole of it, in the middle of the northern hemisphere's summer. The seasons are not caused by distance. They are caused by a lean, the twenty-three-degree tilt of the planet's axis, and everything that follows in this book follows from that single geometric fact: the tilt that points the north pole at the Sun for half the year and away for the other half, that stretches the summer days and shortens the winter ones, that stands the Sun high in June and low in December, that divides the planet into a temperate belt of four seasons, a tropical belt of two, and polar caps of midnight suns and noon darknesses.

This book is the biography of that lean. It follows the long human project of understanding what the seasons are, from the farmers who watched the sky for planting signs and the priests who built calendars to the Greek geometers who measured the tilt with shadows, the astronomers who tamed the year's length into the great calendars of the world, and the modern scientists who watch the seasons from orbit and find them shifting. It explains the physics plainly, the geometry of sunbeams and the slow arithmetic of daylight, and it explores the consequences: the migrating heat of the tropics, the delayed warmth of July after the June solstice, the months-long days of the polar summer, and the seasons of other worlds, where tilts of other sizes make calendars that would be unrecognizable to us.

The story deserves telling because it corrects a famous error on the way. Most people, asked why summer is warm, say the Earth is closer to the Sun, and the answer is not merely wrong but backwards, wrong in a way that survived most of the reader's own schooling. The true explanation, the tilted axis and the geometry of light, is more beautiful than the wrong one, and it explains everything the wrong one cannot: why the two hemispheres have opposite seasons, why the tropics have no summer and winter at all, why the poles have one long day and one long night, and why the dates of the seasons, the solstices and equinoxes, are properties of the sky rather than of any particular weather.

The book's plan follows the series' established shape. The opening four chapters lay the foundations: the history of humanity's long figuring-out of the seasons, the plain physics of the tilt, the characteristics of the seasonal year, and the geometry and heat arithmetic behind it all. The middle chapters cover observation, from shadow-clocks to satellites, and the wider contexts, cultural and planetary. The later chapters carry the story to the present: the modern science of the seasons, the open questions, the future of the Earth's lean, and the curiosities, perihelion in January, the figure-eight of the noon Sun, the equinox that is not quite equal, that make the subject a pleasure to know. No special knowledge is needed to follow it, only the ordinary curiosity of a person who has lived through a few dozen springs and wondered, at least once, what exactly one is.

A note on the book's scope, since the word "seasons" carries more than one meaning in the world's libraries. The seasons of this volume are the Earth's, the astronomical and climatic year of spring, summer, autumn, and winter, and the reader looking for the seasons of the musical or artistic tradition will find them acknowledged in the context chapter and left to their own biographies; the world's catalogues contain more than one book with a name like this one, and more than one of them is about a set of concertos or a songbook. This book is about the planet's year: the tilt that causes it, the history that measured it, the calendars that tamed it, and the weather, ice, oceans, and living things that answer it. Where the seasons' cultural presence, the festivals, the metaphors, the harvest and the midwinter feast, enters the story, it enters as evidence of how much the sky's year has always meant, which is part of the subject and worth its chapter; but the spine of the book is geometry, and the reader finishes it knowing the cause of the year's turning, not merely its moods.

The tilt's number deserves a moment before the book begins, since one number carries the whole subject. Twenty-three and a half degrees, the angle between the Earth's axis and the perpendicular to its orbit, is small enough to seem accidental, a fifth of a right angle, a lean the reader could mime with a slight bow of the head, and yet the entire seasonal year, the world's calendars and harvests and festivals and migrations, hangs from it. If the lean were zero the seasons would vanish, every latitude keeping one weather the year round; if it were doubled the seasons would grow extreme, the tropics reaching into the temperate latitudes' swing; at ninety degrees, the Uranian extreme, the year would become a light-and-dark cycle stranger than any season. The Earth's actual lean, middling and steady, makes the world this book describes, and the number's modesty is part of its wonder: a slight

tilt, held for billions of years, has written every spring the planet has ever had.

Chapter 1: The Discovery of The Seasons

The seasons were used before they were understood, and the distance between the two is the subject's first story.

Humanity's oldest seasonal knowledge was practical, the noticing that certain sky events and certain Earth events arrive together. The Nile floods when certain stars rise before dawn, the rains come when the Pleiades appear, the cold follows the Sun's low path, and the earliest calendars, whatever else they were, were records of such pairings, sky-borne signs for Earth-borne work. The old farming texts of the ancient world, the Greek poet's almanac of when to plough and when to sail, the Egyptian flood-timings, the Babylonian star-lists, all encode a working model of the year in which the sky is the clock and the seasons are the dial. The builders of the great stone monuments of the neolithic world left, in many of them, alignments that make the stones a calendar: the midwinter sunrise caught in a stone corridor, the year's turning marked in architecture, proof that the solstice, the day the Sun stands still and turns back, was watched and weighed important millennia before writing.

The ancients' theoretical understanding, however, began with the geometry. The Greek astronomers, from about the fifth century before the present era, built the first true model of the Sun's yearly path: the Sun, on their picture, rides a great circle around the sky, tilted against the fixed stars, and the seasons are its stations. The year's four key points, the solstices, where the Sun's northing stops and turns, and the equinoxes, where it crosses the middle line, were named and measured; the tilt of the Sun's path, the obliquity of the ecliptic, the angle that makes the whole seasonal drama, was measured by the Greeks with astonishing economy, using nothing but a stick, a shadow, and a knowledge of the year. Eratosthenes, the geometer of Alexandria, measured the tilt from the solstice shadows and had the value right to within a fraction of a degree, an achievement this book will detail in the observation chapter. The essential piece of understanding, that the seasonal year is a geometric fact of the sky, was in place more than two thousand years ago.

The harder step, the one that took the world much longer, was the leap from the Sun's path to the Earth's lean. The ancients' picture, of a Sun circling a central Earth, put the geometry in the wrong place; it worked perfectly for prediction, and the calendars built on it, refined over centuries into the great solar calendars of the ancient and medieval world, kept the solstices and equinoxes on schedule for farming and liturgy alike. But the Copernican rearrangement of the sixteenth century, which put the Sun at the center and the Earth in motion, forced the reinterpretation, and by the seventeenth century the modern understanding had taken shape: the Earth is a tilted spinning ball traveling around the Sun, and the seasonal year is the record of the tilt's changing orientation toward the star. The old geometry of the sky, the tilted circle and its four stations, survived the rearrangement intact, re-interpreted as the Earth's own behavior; the seasons moved from the Sun's journey to the Earth's posture, but the calendar, to the credit of the old astronomers, never had to change at all.

One more historical step completes the discovery. The exact length of the seasonal year, the time from one spring equinox to the next, is not quite the same as other ways of counting a year, and the difference, a matter of minutes, defeated the world's calendars for sixteen centuries. The Julian calendar, the Roman reform that fixed the year at three hundred sixty-five and a quarter days, was true enough, but its small error accumulated, and by the sixteenth century the spring equinox, the Church's own clock for its spring festival, had drifted ten days from the calendar's reckoning. The Gregorian reform of 1582, the calendar the world now uses, was a fix to the length of the seasonal year, and its story, told in the context chapter, is the best proof that the seasons, the equinox in particular, were the single most measured quantity in all of pre-modern science.

The discovery of the seasons, then, is a three-thousand-year collaboration: the farmers and priests who used the pattern, the geometers who measured its geometry, the calendar-makers who tamed its length, and the astronomers who finally explained its cause. The reader who steps out at the June solstice and watches the longest sunset is standing at the end of that collaboration, watching the same turning the stone monuments were built for, with the cause, at last, fully in hand.

The Chinese and Indian astronomical traditions deserve their paragraph in the discovery story, for they ran the same sky-clock project on their own terms. The Chinese calendar-makers, from antiquity through the imperial era, kept the year's measurement a matter of the state, the solstices and equinoxes fixed by the official astronomers with gnomons and clepsydras, the calendar promulgated by the throne and adjusted by its mathematicians, a tradition of seasonal measurement as old and as precise as the Mediterranean one; the Indian astronomical tradition, likewise, carried its own solar reckoning, the sidereal and tropical years distinguished in its classical texts, and the festival calendar of the subcontinent still shows the mixture, solar solstices and lunar months sharing the

ritual year. The discovery of the seasons, the reader should understand, was humanity's project, not one civilization's, and the tilt's geometry was measured by every tradition that built a calendar worth the name, each with its own instruments and its own share of the answer.

The historical measurement of the tilt deserves its own record here, before the observation chapter takes up the instruments, because it is the discovery story's purest episode. The tilt's size, the obliquity, shows itself in the sky directly: the Sun's noon height swings between two extremes through the year, and the swing, twice the tilt, is readable by anyone with a stick and a year of patience. The ancients' recorded values, from the Babylonian and Greek traditions onward, cluster around the correct figure, and the small drift in the recorded values through antiquity, a slow decrease over the centuries, is real, the tilt itself easing by a fraction of a degree across the millennia, so the old astronomers' slightly larger numbers were correct for their era; the modern value, about twenty-three and a half degrees, is the current reading of a quantity the oldest sky-watch measured with creditable accuracy, and the reader who wants to feel the continuity of the science may hold the two facts together, that the tilt has been known within a degree for over two thousand years, and that its exact drift is one of the quantities the modern astronomers still chart.

The discovery story's honest end deserves recording too, the piece of the understanding that arrived last and most reluctantly: the public's. The professional understanding, the tilted Earth in orbit, has been settled science for three centuries, and yet the distance explanation, the near-Sun summer, has survived the whole of modern schooling as the world's favorite guess, and the education researchers who study the error find it entrenched early and corrected slowly. The discovery of the seasons is thus complete in the journals and incomplete in the street, a gap this book's final chapter addresses directly, and the reader should take the historical lesson from it: the seasons were used for millennia, explained for centuries, and are still, in the strict sense, being discovered, one reader at a time, which is why a book like this one is written at all.